Harnessing Machine Learning for Early Detection of Alzheimer's Disease

Ankita Tyagi[1]
Amity School of Engineering and Technology, Amity University
Uttar Pradesh, Noida India
ankitatyagi1313@gmail.com

Eashita Jhingran[2]
Amity School of Engineering and Technology, Amity University
Uttar Pradesh, Noida India
eashitajhingran@gmail.com

Dr Bhupendra Singh[3]
Amity School of Engineering and Technology, Amity University
Uttar Pradesh, Noida India
bsingh5@amity.edu

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***Abstract*— Alzheimer's Disease (AD) is a degenerative neurological condition that results in the deterioration of cognitive function. Early non-invasive detection is crucial for implementing timely interventions and slowing disease progression. In this research, we aimed to analyse the predictivity for classifying AD stages by integrating neuroimaging data—diffusion tensor imaging (DTI) and structural MRI (sMRI)—with clinical biomarkers. In this paper Fractional Anisotropy (FA) scans, derived from DTI, denoting water molecule diffusion in brain tissue was assessed, particularly in identifying early microstructural changes during the mild cognitive impairment (MCI) stage of AD. To progressively monitor the structural pathology of AD in patients, OASIS sMRI dataset was used. Convolutional neural networks (CNNs) like ResNet 3D and LeNET were applied to analyse FA and sMRI data, achieving a significant accuracy score. In this proposed work, correlation and permutation analysis was also applied on the blood biomarker data to extract relevant features. Random Forest classifier was then used on these biomarkers. The highly correlated biomarkers were later used to obtain a significant impact in identifying the various stages of Alzheimer's progression.**

Keywords— ***Alzheimer's disease, Fractional Anisotropy (FA), Diffusion Tensor Imaging (DTI), Biomarkers, Magnetic Resonance Imaging (MRI), Early diagnosis***

I. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that affects millions of people worldwide, particularly the elderly. As the global population ages, the number of AD cases is expected to rise significantly, putting immense pressure on healthcare systems. AD is marked by a steady decline in cognitive function and the loss of functional abilities, and its pathological hallmarks, such as amyloid plaques and neurofibrillary tangles, often only become detectable after substantial and irreversible brain damage has occurred. This highlights the pressing need for early detection, as timely interventions may help slow disease progression and improve quality of life for those affected.

Traditional diagnostic methods, such as clinical assessments and cognitive tests, often fail to detect AD in its earliest stages, when intervention would be most effective. To address this limitation, advanced neuroimaging techniques have become essential tools in AD research and diagnosis. Diffusion tensor imaging (DTI), for instance, measures the diffusion of water molecules in brain tissues, providing valuable insights into white matter integrity through metrics like Fractional Anisotropy (FA). FA reflects early microstructural changes in white matter, which are among the first signs of AD. Similarly, structural MRI (sMRI) is crucial in assessing gray matter atrophy, especially in regions like the hippocampus, which is particularly vulnerable to early damage in AD. Recent studies have increasingly highlighted the benefits of integrating multiple imaging modalities, such as FA from DTI and sMRI data, to enhance diagnostic accuracy. Combining MRI and genetic data with machine learning models has also led to diagnostic accuracy rates as high as 90%, with models like ADD-NET demonstrating strong performance by optimizing multimodal data [1]. These multimodal approaches have shown great promise in distinguishing between AD, mild cognitive impairment (MCI), and cognitively normal individuals, demonstrating a more refined methodology for early detection and intervention.

Recent advances in machine learning have added another layer of sophistication to the use of multimodal data in AD diagnosis. For example, research demonstrates the potential of deep learning models applied to brain MRI images, significantly improving detection accuracy and offering a robust framework for early diagnosis [2]. CNN-based models have further advanced this field, achieving classification accuracies of over 90% in diagnosing Alzheimer's, thus illustrating how the integration of machine learning with MRI data can enhance early detection capabilities [3]. The effectiveness of these convolutional neural networks (CNNs) in identifying subtle structural changes in the brain has also been consistently emphasized.

Equally important in advancing AD diagnostics are ML models utilizing novel biomarkers which can reliably distinguish between normal control and demented individuals [5]. Moreover, the strategic selection of biomarkers, particularly those associated with inflammation, oxidative stress, and metabolic processes, have gained significant attention for their potential to increase the scope of sensitivity of early detection models [6]. Further research has explored the

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links between mood disorders, such as depression, and AD progression, revealing plasma markers of neurodegeneration as crucial indicators that may inform both diagnosis and disease monitoring [7]. Large-scale studies, including the Framingham Heart Study, have also reinforced the significance of these blood-based biomarkers, using machine learning techniques to identify key biomarkers associated with dementia, further highlighting their value in the early detection of AD [8]. Blood-based biomarkers (BBMs), such as plasma Aβ42/Aβ40, p-tau, and NfL, when integrated with machine learning techniques, offer enhanced diagnostic accuracy and reduce reliance on costly methods like PET and CSF tests [9].

In addition to MRI and blood biomarkers, FA scans from DTI have emerged as a critical component in AD research. Numerous studies have demonstrated that FA measures can effectively distinguish between AD patients and healthy controls by analysing white matter integrity, which notably decreases as the disease progresses [10]. Machine learning methods have proven invaluable in analysing these relationships, revealing complex interactions between white matter integrity and cognitive function. Moreover, advanced models incorporating self-attention mechanisms with FA data have achieved high classification accuracy, particularly when focusing on brain regions specifically impacted by AD [11]. These findings, combined with sophisticated classification algorithms, further solidify FA's role in early AD detection, offering significant clinical potential [12].

Building on these advancements, our proposed work aims to bridge the gaps in AD detection by developing a more comprehensive and integrated-multimodal approach. By combining structural MRI data, blood biomarkers, and FA measures with cutting-edge machine learning techniques, we seek to improve diagnostic accuracy while gaining a more extensive understanding of AD pathology. By synthesizing insights from these diverse data sources, we aim to create predictive models that are both robust and precise, enabling more effective early detection of AD. Preliminary results from our experiments have already shown promising classification accuracies, highlighting the potential of this multimodal approach to improve early intervention strategies and ultimately enhance clinical outcomes for individuals affected by Alzheimer's disease.

The remainder of this paper is structured as follows: Section 2 discusses the datasets utilised in this work and the methodology employed which spanned over the pre-processing stages, model deployment and model evaluation. The experimental results are discussed in Section 3 followed by a discussion of the findings in context of our research problem and the existing literature, and finally Section 4 concludes the paper, also outlining future research discussions.

II. METHODOLOGY

A. Dataset

1) *OASIS Dataset*: The Open Access Series of Imaging Studies (OASIS) dataset, is a comprehensive resource widely used in Alzheimer's disease (AD) research. Established to facilitate the study of brain aging and neurodegenerative diseases, OASIS provides a rich collection of cross-sectional and longitudinal MRI data from a diverse cohort of subjects, including both healthy controls/non-demented(ND) and individuals diagnosed with various stages of Alzheimer's disease—Very Mild Demented (VMD), Mild Demented (MiD) and Moderate Demented (MoD).

Our dataset comprised of 80,000 brain MRI images, which were used to study Alzheimer's disease progression. Original NIfTI files (.nii) files were converted into 2D image formats like .jpeg using FSL and were made available through a GitHub repository. For neural network training, the brain scans were sliced into 2D images, selecting slices 100 to 160 along the z-axis for 461 patients. The OASIS dataset included T1-weighted structural MRI scans, along with associated clinical assessments such as the Clinical Dementia Rating (CDR) and Mini-Mental State Examination (MMSE) scores. Based on Clinical Dementia Rating (CDR) values, patients were classified into four stages: demented, very mild demented, mild demented, and non-demented. The final dataset, processed into JPEG format, totalled up to 1.3 GB, enabling robust Alzheimer’s detection analysis.

2) *Biomarker:* The biomarker dataset utilized in this research integrates a diverse range of clinical, cognitive, neuroimaging, and biomarker information, providing a comprehensive view of Alzheimer's disease pathology. Key biomarkers such as Brain-Derived Neurotrophic Factor (BDNF), Interleukins (IL1, IL6), and Tumor Necrosis Factor (TNF) are crucial indicators of neuroinflammation, toxic protein secretion and oxidative stress, which play pivotal roles in disease progression. In addition to these biomarkers, the dataset includes neuroimaging metrics such as hippocampal volume, cortical thickness, and ventricle size, which help in identifying structural brain changes linked to Alzheimer's disease. Cognitive assessments, including the Mini-Mental State Examination (MMSE), Cambridge Cognitive Examination (CAMCOG), and various other tests which are used to evaluate patients' memory, attention, and executive function were also used in model training. The dataset also includes information on medications taken, such as anti-hypertensives, antidepressants, and anticholinergics, which can influence cognitive outcomes. Other clinical factors such as age, education, and comorbidities (e.g., hypertension, diabetes) further contextualize individual health profiles. This multidimensional dataset serves as a robust foundation for developing machine learning models aimed at improving early detection and understanding of Alzheimer’s disease progression.

3) *Fractional Anisotropy (FA):* FA Scans, derived from Diffusion Tensor Imaging (DTI) which were obtained from the portal of EU Open Research Repository, measure the diffusion of water molecules in the brain to provide insights into white matter (WM) integrity. High FA values are generally observed in healthy white matter, where water diffusion is highly directional along axonal fibres, while lower FA values indicate white matter degradation, which is often associated with pathological processes like Alzheimer’s disease (AD).

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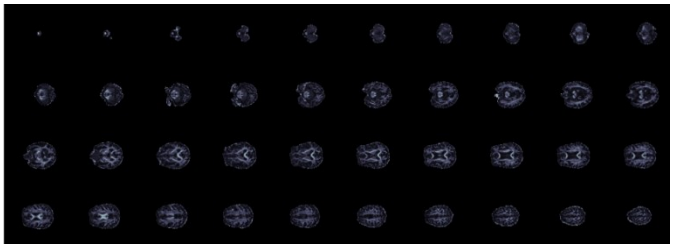

Fig. 1. *Fractional Anisotropy 40 longitudinal scan slices(128x128x45)*

B. *Preprocessing*

Preprocessing of the neuroimaging data involved several steps. In this paper, first, the FA data was motion-corrected and normalized to ensure consistency across subjects.

Next, the FA values were extracted from specific regions of interest (ROIs) using standard anatomical templates. These ROIs were chosen based on previous studies indicating their vulnerability to early AD-related changes.

For the sMRI data, Voxel Based Morphometry(VBM) was used to measure gray matter volume in the hippocampus, entorhinal cortex, and other cortical areas. The images were normalized to the Montreal Neurological Institute (MNI) template, and gray matter volumes were extracted for statistical analysis.

The biomarker data was pre-processed by handling missing values through imputation. For categorical data (e.g., gender, diagnosis), label encoding was applied. A correlation analysis was performed to identify the most relevant biomarkers for classification tasks. The top 20 biomarkers were selected based on their correlation with cognitive scores and AD diagnosis.

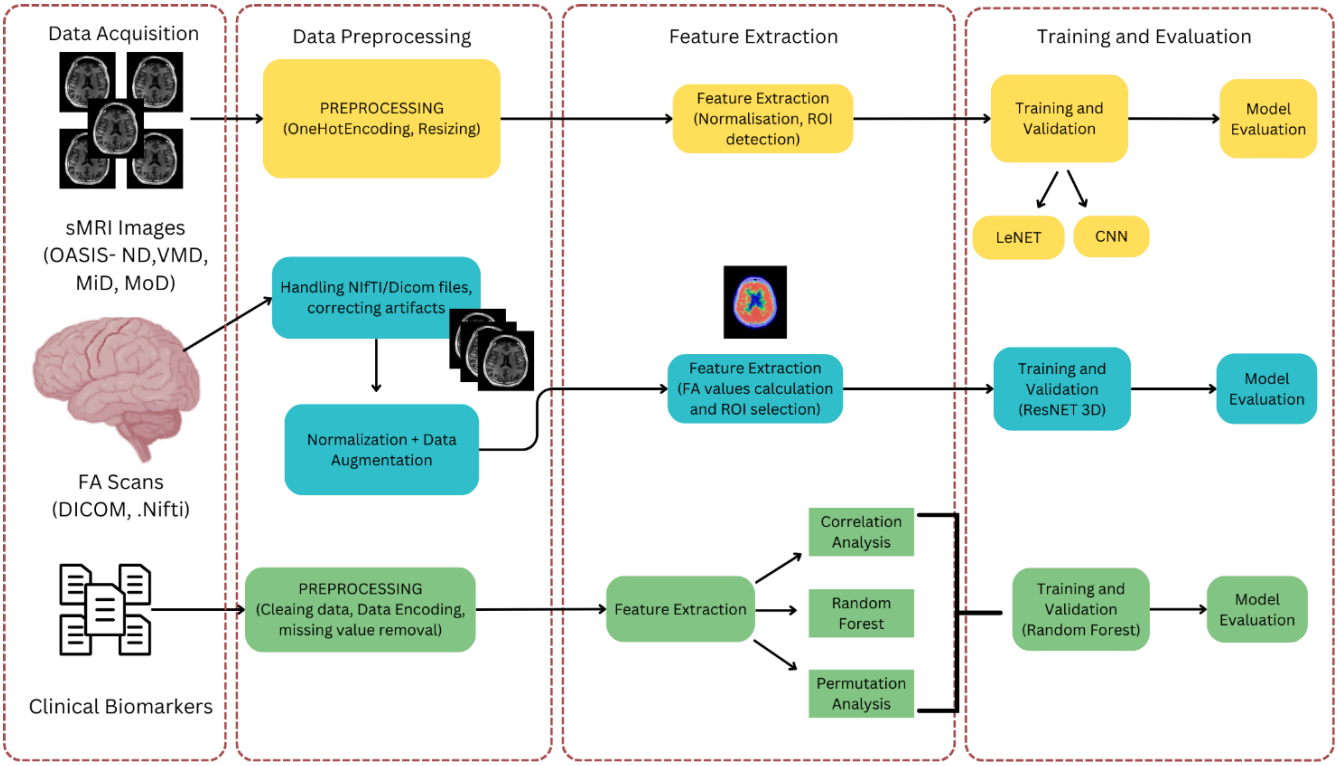

Fig. 2 *Research Methodology*

C. *Model Deployment*

The machine learning model employed in this paper processed extensive 3D imaging data from Fractional Anisotropy (FA) scans, 2D structural MRI (sMRI) from the OASIS dataset, and tabulated clinical biomarkers to classify Alzheimer's disease (AD) stages effectively. The process consisted of data acquisition, model selection, training and validation, and model optimization as represented in Fig. 2.

CNN models like ResNeT 3D [14] and LeNet were applied to the neuroimaging data, specifically FA and sMRI. The CNN architecture included several convolutional layers followed by max pooling and fully connected layers as shown in Fig. 3 below. This architecture allowed the model to capture spatial hierarchies in the data and detect subtle differences in brain structure degeneration as AD progresses.

For the biomarker data, Permutation Analysis and Correlation analysis were employed for feature extraction. Random Forest classifier was also tested for its robustness in handling high-dimensional data and its ability to rank feature importance. The final classifier was trained on the selected biomarkers and was evaluated based on accuracy, precision,

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recall, and AUC-ROC (Area under the Receiver Operating Characteristic Curve).

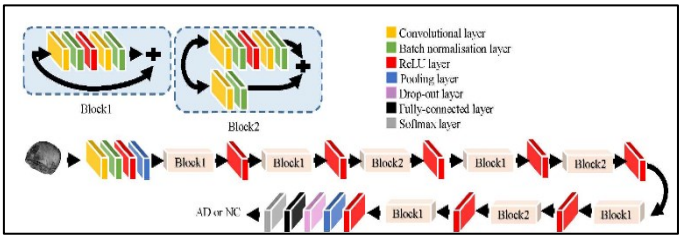

Fig.3. Basic structures of the ResNet 3D

The datasets were divided into training, validation, and test sets using an 80-20 split. The models were trained on the training set, validated on a separate validation set, and then tested on unseen data to evaluate their generalizability. Following the deployment, hyperparameter tuning was conducted to determine the optimal parameters for each model.

The final model, selected based on the highest cross-validation accuracy and generalization performance, integrated both imaging and clinical data to improve the early detection of AD.

D. Model Evaluation

The optimized models were evaluated on held-out test sets to assess their performance in distinguishing between AD progression stages. Metrics such as accuracy, precision, recall, F1-score, and AUC-ROC were calculated to provide a comprehensive evaluation of the model's performance. These metrics are defined as:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \text{Precision} = \frac{TP}{TP+FP}, \text{Sensitivity} = \frac{TP}{TP+FN},$$

$$\text{F1 score} = \frac{2TP}{2TP+FP+FN}$$

Where FP=False Positive, FN= False Negative, TP= True Positive, and TN= True Negative.

The integration of multiple imaging modalities and biomarkers significantly enhanced the classification approach, highlighting the potential of a multi-modal technique for early Alzheimer's diagnosis.

III. RESULTS AND DISCUSSION

A. Results

The evaluation of the individual models for Fractional Anisotropy (FA), structural MRI (sMRI), and biomarker data was carried out using their respective test datasets. Each modality was processed and modelled independently with the goal of future integration. Although a robust integrated model is still in progress, the initial results from individual models show promising accuracy, underscoring the potential of a multi-modal approach.

(1) Fractional Anisotropy (FA) Model: The FA model assessed white matter integrity in brain tissues which degrade overtime in AD patients. Significant differences in FA values across the Alzheimer's Disease (AD) progressive stages were identified, particularly in the cingulum bundle, uncinate fasciculus, and corpus callosum. These reductions in FA values suggest early white matter degeneration, a less utilised early marker of Alzheimer's. The FA model achieved 82.5% classification accuracy, effectively identifying early-stage degeneration, especially in the transition from CN to MCI.

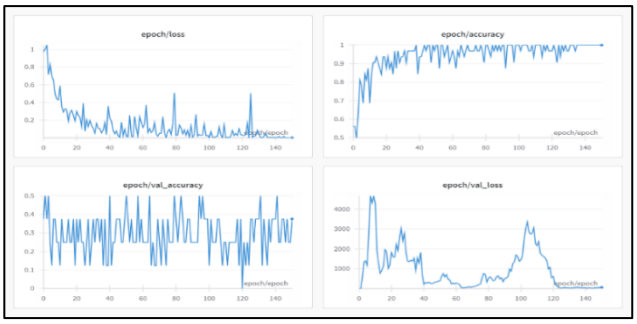

Fig. 4. FA Scans Model training metrics using ResNet 3D for 150 epochs.

In the Fig. 4 training curves show a steady decrease in loss and a gradual increase in accuracy, with the model approaching high performance after around 40 epochs and validation loss increased after an initial drop, indicating potential overfitting hence why Early Stopping was employed to curb the issue.

(2) Gray Matter (GM) Model from Structural MRI: The sMRI model focused on measuring gray matter atrophy in regions such as the hippocampus and entorhinal cortex. Voxel-Based Morphometry (VBM) revealed significant atrophy in MoD and MiD patients compared to ND subjects. The CNN models like LeNet and a custom variation achieved accuracies of 99.3% and 99.7% respectively in classifying ND, VMD, MiD, and MoD subjects, confirming the importance of sMRI in detecting structural changes during later stages of AD. The performance metric curves are mentioned in Fig. 6 below.

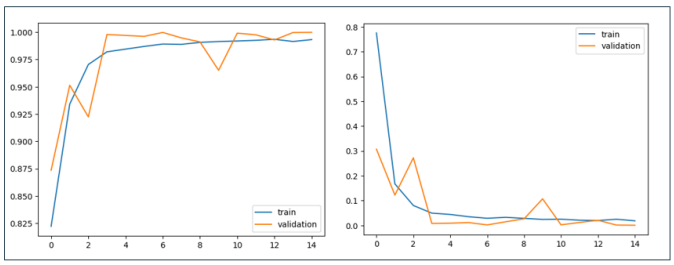

Fig. 5. (a)Training Accuracy and (b) Training Loss Curves of the LeNET Model Training and Validation.

In the Fig. 5, training curves show rapid accuracy improvement within the first few epochs, reaching near-perfect performance by epoch 5, while the loss converges quickly. The alignment of training and validation metrics suggests effective learning without signs of overfitting.

(3) Biomarker Model: The biomarker model utilized clinical data, including neuroinflammatory markers (such as MMSE, Hippocampus, etc), which were pre-processed and standardized. The model achieved 90% accuracy,

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demonstrating the significant role of biomarkers in distinguishing between CN, MCI, and AD stages. The results highlight the potential of modes for dianosis using biomarker data.

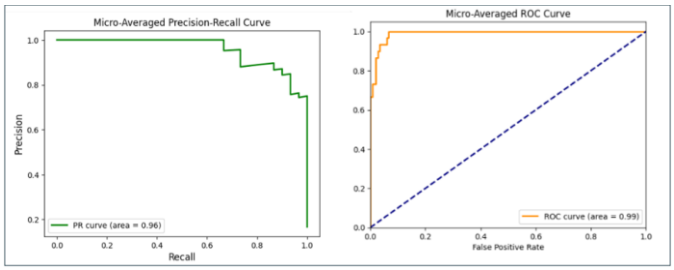

Fig. 6. (a) Precision and (b)AUC-ROC curves of the Random Forest Model Training and Validation.

Fig. 6 shows the Precision and AUC-ROC curve depecting the model that achieved 90% accuracy, demonstrating the significant role of selected biomarkers in distinguishing between CN, MCI, and AD stages.

TABLE I. PERFORMANCE OF OUR PROPOSED MODELS

Datasets	**Performance**	
	Proposed Modality	***Accuracy***
OASIS sMRI	CNN	**99.7**
	LeNet	**99.3**
FA Scans	ResNet18 3D	**82.5**
Biomarkers	Random Forest	**90**

TABLE II. PERFORMANCE OF EXISTING STATE-OF-THE-ART MODELS

Datasets	**Performance**		
	Author and Year	***Existing Modality***	***Accuracy***
OASIS sMRI	Battineni et al. (2021) [17]	Gradient boosting	97.58%,
	Islam J. et al, 2018[16]	CNN	93%
FA Scans	Yida Qu et al, 2021[18]	SVM, XGBoost, L.R. etc	80.47%
	Y. Sang et al, 2024[19]	Graph Conv. Network	78.6%
Biomarkers	Popuri et al., 2020 [15]	ensemble-learning	AUC: 0.81(6 Months), 0.73(7 Yrs)
	Jo et al., 2020[13]	CNN	90.8%

The above Table 1 and Table 2 compare the performance accuracies of our models against the existing modalities, thus proving that our approaches are significantly detecting plausible diagnostics.

B. Discussion

The high accuracies obtained emphasize the significance of having a comprehensive neuroimaging and biomarker data basket for improved classification of early-stage Alzheimer's. Each model—FA, sMRI, and biomarker-based—offers valuable insights. FA were effective at detecting early white matter degeneration, while sMRI captured gray matter atrophy in later stages. Biomarkers provided more insights into underlying biological processes such as inflammation, toxic proteins and oxidative stress, complementing our imaging data.

A multi-modal approach hence is crucial for accurately diagnosing AD. This is understood because while FA reveals early white matter changes, it lacks specificity for tracking gray matter atrophy. Conversely, sMRI provides detailed structural information but at a much later stage where patients are usually symptomatic and miss the golden period of reverting back from the damage. It misses early white matter abnormalities. Biomarkers, though effective for detecting underlying biological processes, cannot directly reveal structural changes in the brain. Therefore, integrating these modalities becomes necessary for a complete and accurate early diagnosis, and we have laid the groundwork for enabling a more effective diagnostic approach.

IV. CONCLUSION AND FUTURE WORK

A. Conclusion

The proposed work has highlighted the significant potential of multimodal integration for improving the early detection of Alzheimer's disease (AD). By combining advanced neuroimaging techniques—such as FA metrics derived from DTI, which reflect white matter integrity, and sMRI-based measurements of gray matter volume—with blood biomarker analysis, and applying machine learning models, significant improvements in diagnostic accuracy can be achieved. Specifically, LeNET and a Customized CNN model demonstrated remarkable accuracies of 99.3% and 99.7%, respectively, in analyzing MRI data, while the Random Forest classifier achieved an accuracy of 90% in biomarker data analysis. Additionally, ResNet 3D contributed to classifying various stages of AD with an accuracy of 82.5%, demonstrating its utility in capturing disease progression patterns at micro level.

While the full potential for improving diagnostic accuracy remains to be fully explored, these results demonstrate the feasibility of integrating multimodal data types, offering a promising path forward for AD research and clinical applications. By leveraging this combination of imaging and biomarker data, this study provides a more comprehensive framework for understanding AD progression. Furthermore, the integration of these modalities enhances diagnostic precision and reveals patterns that might otherwise remain

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undetected when using single modalities improving patient care in the long term.

B. Future Work

(1) Integration of Multi-Modal Data: Future research will focus on developing a unified model combining FA, sMRI, and biomarker data. This integrated approach is expected to improve accuracy in detecting early-stage AD and monitoring disease progression. Combining structural and biological data will capture the full spectrum of AD pathology and provide deeper insights into the disease.

(2) Expansion of Data Sources: Expanding the dataset, including data from resources such as the Alzheimer's Disease Neuroimaging Initiative (ADNI), will improve model generalizability and robustness. Additionally, incorporating more biomarkers data such could improve classification accuracy and facilitate better stratification of AD subtypes.

(3) Improving Model Interpretability: One of the key challenges in deep learning is the interpretability of complex models. Techniques like Gradient-weighted Class Activation Mapping will be explored to increase transparency, highlighting that the brain regions or biomarkers influence the model's predictions. Improving interpretability is crucial for clinical adoption.

(4) Development of a User-Friendly Prediction Support System: A practical Alzheimer's Prediction Support System (APSS) will be developed to assist clinicians in early diagnosis. This system will integrate neuroimaging and biomarker data for real-time analysis and decision-making. Designed for ease of use, the APSS will offer actionable insights and diagnosis results, ultimately improving patient outcomes through early intervention.

(5) Incorporating Advanced Models: Further improvements in accuracy can be achieved by exploring hybrid models that combine traditional machine learning algorithms with neural networks. Integrating Random Forests with deep learning could leverage the strengths of both approaches to handle structured and unstructured data more effectively.

In conclusion, this paper demonstrates the potential for machine learning models that integrate neuroimaging and biomarkers to enhance early detection of AD. The findings lay the groundwork for future advancements that could transform clinical practices by a multi modal approach, improving diagnosis, and enabling earlier interventions, ultimately enhancing patient outcomes.

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